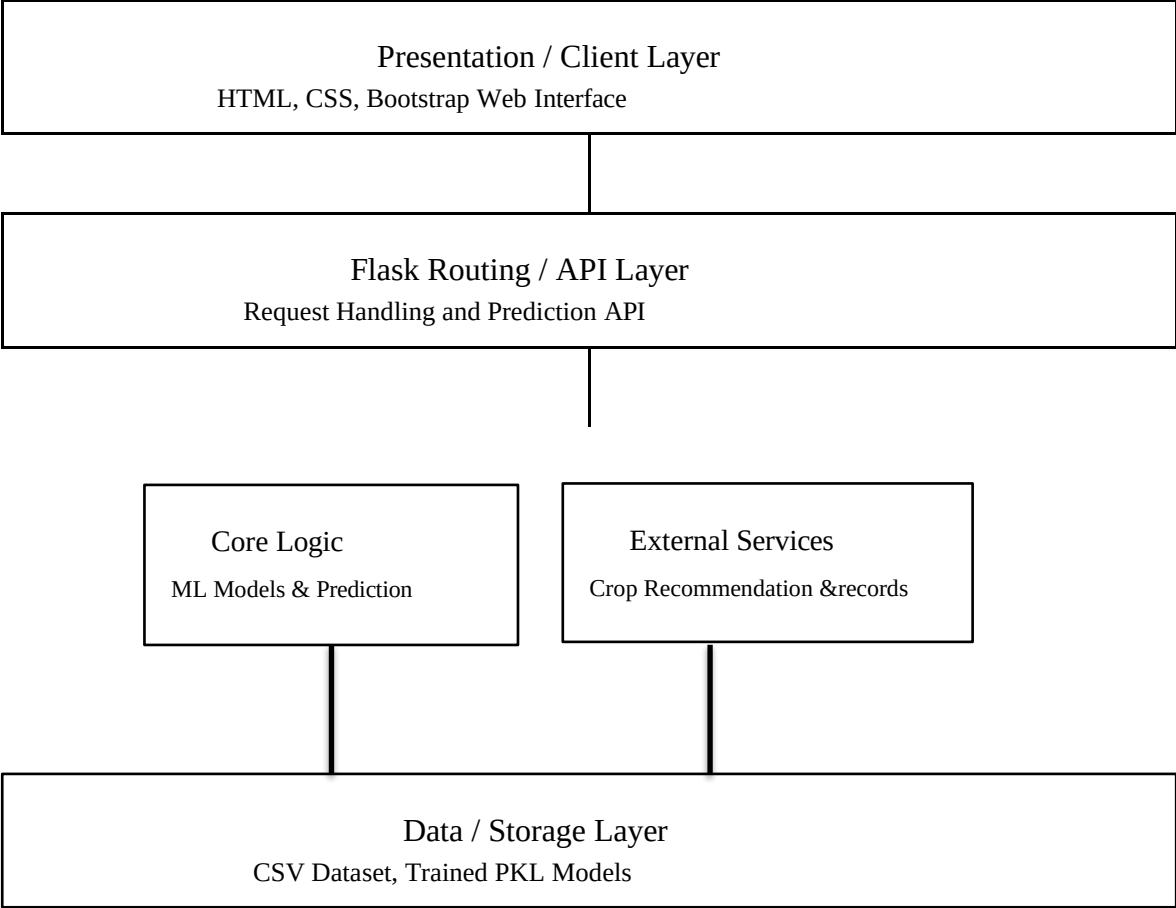


Solution Architecture

Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects soil and environmental data from users and displays crop recommendations.	HTML, CSS, Bootstrap
API Layer	Handles requests between the user interface and the Machine Learning model.	Flask
Core Logic Layer	Processes input data and predicts the most suitable crop using Machine Learning.	Scikit-learn, Pandas, NumPy
Data / Storage Layer	Stores the crop dataset and trained Machine Learning model.	CSV Dataset, Pickle (.pkl)